

CHE 230: Molecular Systems in Chemical Engineering Homework 3

Due: 4/01/2026

CHE 230 Time: 11:59pm

1. a. Derive the lever rule. (5 points)

Rule of mass balance

Let C_0 = overall composition
Let C_α = composition of phase α
Let C_β = composition of phase β
Let W_α, W_β = mass fractions

$$C_0 = W_\alpha C_\alpha + W_\beta C_\beta$$

$$W_\alpha + W_\beta = 1$$

$$W_\beta = 1 - W_\alpha$$

$$C_0 = W_\alpha C_\alpha + (1 - W_\alpha) C_\beta$$

$$= W_\alpha C_\alpha + C_\beta - W_\alpha C_\beta$$

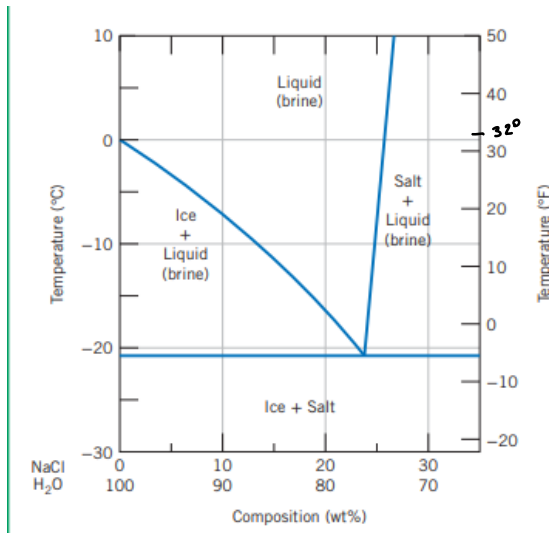
$$C_0 = W_\alpha (C_\alpha - C_\beta) + C_\beta$$

$$\frac{C_0 - C_\beta}{C_\alpha - C_\beta} = W_\alpha$$

So

$$W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha}$$

b. The following is a portion of the H_2O – NaCl phase diagram: (5 points)



(i) Using this diagram, briefly explain how spreading salt on ice that is at a temperature below 0°C (32°F) can cause the ice to melt.

(ii) At what temperature is salt no longer useful in causing ice to melt?

(i) As the composition of NaCl rises this shifts the system to a region of liquid phase at the same temperature. So the more NaCl added, the more the system shifts to those liquid or brine phase.

(ii) At -21°C the system stays in a solid composition of salt & ice

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2. 90 wt% Ag-10 wt% Cu alloy is heated to a temperature within the β + liquid phase region. If the composition of the liquid phase is 85 wt% Ag, determine: (14 points)

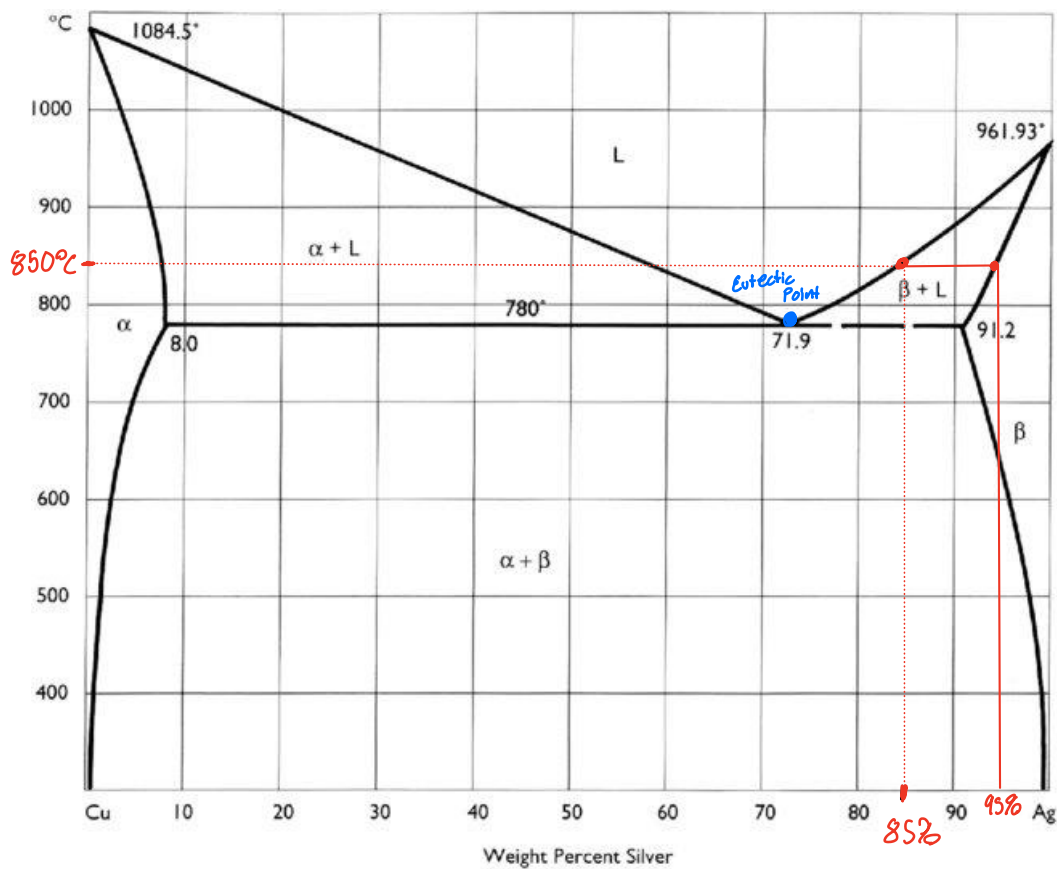
(a) The temperature of the alloy: 850°C

(b) The composition of the β phase: $95\text{ wt}\% \text{ Ag}$

(c) The mass fractions of both phases

$$W_{\beta} = \frac{90 - 85}{94 - 85} \approx 0.5$$

$$W_L = \frac{94 - 90}{94 - 85} \approx 0.5$$



3. a. Explain the phenomenon of temperature change from a molecular perspective and why it is that some materials have higher heat capacity than others (6 points).

From a molecular perspective temperature can be perceived as the average kinetic energy. When temperature rises molecules begin to move in a multitude of ways. However some materials are able to store that energy in more ways than materials with a lower heat capacity. These ways of storing energy include rotations, vibrations, bond stretches and other intermolecular motions.

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- b. Explain why a brass lid ring on a glass canning jar loosens when heated (4 points).

The lid loosens due to the increase in the movement of atoms which cause atoms to move slightly apart. The brass will expand slightly more than the glass which makes it loose.

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- c. A brass rod is to be used in an application requiring its ends to be held rigid. If the rod is stress-free at room temperature (20°C or 68°F), what is the maximum temperature to which the rod may be heated without exceeding a compressive stress of 172 MPa (25,000 psi)?

$$= 172 \times 10^6 \text{ Pa}$$

Assume the following properties for brass:

- Modulus of elasticity (E) = 100 GPa (14.6×10^6 psi) = 100×10^9 Pa
- Linear coefficient of thermal expansion (α) = $20.0 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ ($11.1 \times 10^{-6} \text{ } ^\circ\text{F}^{-1}$) (8 points)

$$\sigma = E\alpha\Delta T$$

$$\Delta T = \frac{\sigma}{E\alpha} = \frac{172 \times 10^6 \text{ Pa}}{(100 \times 10^9 \text{ Pa})(20 \times 10^{-6} \text{ } ^\circ\text{C}^{-1})} = 86^\circ\text{C} + 20^\circ\text{C} = 106^\circ\text{C} = T_{\text{max}}$$

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4. An aluminum wire 10 m (32.8 ft) long is cooled from 38°C to -1°C (100°F to 30°F). How much change in length will it experience? (6 points)

$$\Delta L = \alpha L \Delta T \quad \text{Al } \alpha = 23 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$$

$$L = 10 \text{ m}$$

$$\Delta T = -1 - 38 = -39^\circ\text{C}$$

$$\begin{aligned} \Delta L &= (23 \times 10^{-6} \text{ } ^\circ\text{C}^{-1})(-39^\circ\text{C})(10 \text{ m}) \\ &= -8.97 \times 10^{-3} \text{ m} \end{aligned}$$

✓ 6

5. Compute the electrical conductivity of a cylindrical silicon specimen 5.1 mm (0.2 in.) in diameter and 51 mm (2 in.) in length in which a current of 0.1 A passes in an axial direction. A voltage of 12.5 V is measured across two probes that are separated by 38 mm (1.5 in.).

- (b) Compute the resistance over the entire 51 mm (2 in.) of the specimen (8 points).

$$R = \frac{V}{I} = \frac{12.5}{0.1} = 125 \quad \text{d) } \sigma = \frac{1}{\rho} = \frac{1}{0.066} = 15.01 \text{ } \mu\text{S/m}$$

$$\begin{aligned} A &= \pi r^2 \\ &= \pi (2.55 \times 10^{-3})^2 \\ &= 2.04 \times 10^{-5} \text{ m}^2 \end{aligned}$$

$$R_t = R \cdot \frac{L_{tot}}{L_{meas}}$$

$$\begin{aligned} R_{tot} &= 125 \cdot \frac{0.051}{0.038} \\ &= 168 \text{ } \mu\text{S} \end{aligned}$$

$$\begin{aligned} \rho &= \frac{RA}{L} \\ &= \frac{125(2.04 \times 10^{-5} \text{ m})}{0.038} \\ &= 0.066 \end{aligned}$$

✓ 8

6. An n-type semiconductor is known to have an electron concentration of $3 \times 10^{18} \text{ m}^{-3}$. If the electron drift velocity is 100 m/s in an electric field of 500 V/m, calculate the conductivity of this material (4 points).

$$\sigma = nq\mu$$

$$\mu = \frac{v_d}{E}$$

$$\sigma = nq \frac{v_d}{E}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$\sigma = (3 \times 10^{18}) (1.6 \times 10^{-19}) \left(\frac{100}{500} \right)$$

$$= 0.0096 \frac{\text{S}}{\text{m}}$$

